

## Annex 162C

(normative)

### MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

#### 162C.1 Overview

This annex defines the Media Dependent Interface (MDI) for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4. The MDI couples the PMD (specified in 162.8 and 162.9) to the cable assembly (specified in 162.11). The PMDs supportable for each MDI connector type are given in Table 162C–1. The SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP are also referred to as multi-lane connectors.

**Table 162C–1—Number of PMDs supportable for each connector type**

| MDI types  | 100GBASE-CR1 | 200GBASE-CR2 | 400GBASE-CR4 | Reference |
|------------|--------------|--------------|--------------|-----------|
| SFP112     | 1            | —            | —            | 162C.2.1  |
| SFP-DD112  | 1,2          | 1            | —            | 162C.2.2  |
| DSFP       | 1, 2         | 1            | —            | 162C.2.3  |
| QSFP112    | 1, 2, 4      | 1, 2         | 1            | 162C.2.4  |
| QSFP-DD800 | 1, 2, 4, 8   | 1, 2, 4      | 1, 2         | 162C.2.5  |
| OSFP       | 1, 2, 4, 8   | 1, 2, 4      | 1, 2         | 162C.2.6  |

For 100GBASE-CR1, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of 162C.2.1 (SFP112), 162C.2.2 (SFP-DD112), 162C.2.3 (DSFP), 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800), or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD.

For 200GBASE-CR2, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 162C.2.2 (SFP-DD112), 162C.2.3 (DSFP), 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800), or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD.

For 400GBASE-CR4, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800) or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD. The cable assembly connector shall be the MDI connector plug. The PMD connector shall be the MDI connector receptacle.

The MDI connector electrical performance is consistent with the signal quality and electrical requirements of 162.9 and 162.10.

An MDI connector type may support one or more PMDs. The assignment of PMD signals to connector signals is specified in Table 162C–2, where as an example 0:DL0n refers to the DL0n signal of the first PMD; see 162.8.1 for signal naming definitions. When an MDI connector is not fully utilized the lower PMD numbers in Table 162C–2 should be used.

**Table 162C–2—PMD to connector signal assignments**

| PMD signal <PMD number>:<PMD signal> |              |              | Connector signal |
|--------------------------------------|--------------|--------------|------------------|
| 100GBASE-CR1                         | 200GBASE-CR2 | 400GBASE-CR4 |                  |
| 0:DL0n                               | 0:DL0n       | 0:DL0n       | DL0n             |
| 0:DL0p                               | 0:DL0p       | 0:DL0p       | DL0p             |
| 1:DL0n                               | 0:DL1n       | 0:DL1n       | DL1n             |
| 1:DL0p                               | 0:DL1p       | 0:DL1p       | DL1p             |
| 2:DL0n                               | 1:DL0n       | 0:DL2n       | DL2n             |
| 2:DL0p                               | 1:DL0p       | 0:DL2p       | DL2p             |
| 3:DL0n                               | 1:DL1n       | 0:DL3n       | DL3n             |
| 3:DL0p                               | 1:DL1p       | 0:DL3p       | DL3p             |
| 4:DL0n                               | 2:DL0n       | 1:DL0n       | DL4n             |
| 4:DL0p                               | 2:DL0p       | 1:DL0p       | DL4p             |
| 5:DL0n                               | 2:DL1n       | 1:DL1n       | DL5n             |
| 5:DL0p                               | 2:DL1p       | 1:DL1p       | DL5p             |
| 6:DL0n                               | 3:DL0n       | 1:DL2n       | DL6n             |
| 6:DL0p                               | 3:DL0p       | 1:DL2p       | DL6p             |
| 7:DL0n                               | 3:DL1n       | 1:DL3n       | DL7n             |
| 7:DL0p                               | 3:DL1p       | 1:DL3p       | DL7p             |
| 0:SL0n                               | 0:SL0n       | 0:SL0n       | SL0n             |
| 0:SL0p                               | 0:SL0p       | 0:SL0p       | SL0p             |
| 1:SL0n                               | 0:SL1n       | 0:SL1n       | SL1n             |
| 1:SL0p                               | 0:SL1p       | 0:SL1p       | SL1p             |
| 2:SL0n                               | 1:SL0n       | 0:SL2n       | SL2n             |
| 2:SL0p                               | 1:SL0p       | 0:SL2p       | SL2p             |
| 3:SL0n                               | 1:SL1n       | 0:SL3n       | SL3n             |
| 3:SL0p                               | 1:SL1p       | 0:SL3p       | SL3p             |
| 4:SL0n                               | 2:SL0n       | 1:SL0n       | SL4n             |
| 4:SL0p                               | 2:SL0p       | 1:SL0p       | SL4p             |
| 5:SL0n                               | 2:SL1n       | 1:SL1n       | SL5n             |

**Table 162C–2—PMD to connector signal assignments (*continued*)**

| PMD signal <PMD number>:<PMD signal> |              |              | Connector signal |
|--------------------------------------|--------------|--------------|------------------|
| 100GBASE-CR1                         | 200GBASE-CR2 | 400GBASE-CR4 |                  |
| 5:SL0p                               | 2:SL1p       | 1:SL1p       | SL5p             |
| 6:SL0n                               | 3:SL0n       | 1:SL2n       | SL6n             |
| 6:SL0p                               | 3:SL0p       | 1:SL2p       | SL6p             |
| 7:SL0n                               | 3:SL1n       | 1:SL3n       | SL7n             |
| 7:SL0p                               | 3:SL1p       | 1:SL3p       | SL7p             |

The MDI connectors shall have data signal and signal ground contact assignments as specified in Table 162C–3, Table 162C–4, and Table 162C–5 for the paired transmit and receive lanes (e.g., (SL0, DL0), (SL1, DL1)).

**Table 162C–3—MDI connector contact mapping for SFP112, SFP-DD112, and DSFP**

| SFP112 | SFP-DD112 | DSFP | Connector signal name | Description                         |
|--------|-----------|------|-----------------------|-------------------------------------|
| 11     | 11        | 11   | GND                   | Ground                              |
| 12     | 12        | 12   | DL0n                  | Receiver inverted data output       |
| 13     | 13        | 13   | DL0p                  | Receiver non-inverted data output   |
| 14     | 14        | 14   | GND                   | Ground                              |
| 17     | 17        | 17   | GND                   | Ground                              |
| 18     | 18        | 18   | SL0p                  | Transmitter non-inverted data input |
| 19     | 19        | 19   | SL0n                  | Transmitter inverted data input     |
| 20     | 20        | 20   | GND                   | Ground                              |
| —      | 31        | 10   | GND                   | Ground                              |
| —      | 32        | 9    | DL1n                  | Receiver inverted data output       |
| —      | 33        | 8    | DL1p                  | Receiver non-inverted data output   |
| —      | 34        | 7    | GND                   | Ground                              |
| —      | 37        | 3    | GND                   | Ground                              |
| —      | 38        | 2    | SL1p                  | Transmitter non-inverted data input |
| —      | 39        | 1    | SL1n                  | Transmitter inverted data input     |
| —      | 40        | 22   | GND                   | Ground                              |

**Table 162C–4—MDI connector contact mapping for QSFP112 and QSFP-DD800**

| QSFP112 | QSFP-DD800 | Connector signal name | Description                         |
|---------|------------|-----------------------|-------------------------------------|
| 1       | 1          | GND                   | Ground                              |
| 2       | 2          | SL1n                  | Transmitter inverted data input     |
| 3       | 3          | SL1p                  | Transmitter non-inverted data input |
| 4       | 4          | GND                   | Ground                              |
| 5       | 5          | SL3n                  | Transmitter inverted data input     |
| 6       | 6          | SL3p                  | Transmitter non-inverted data input |
| 7       | 7          | GND                   | Ground                              |
| 13      | 13         | GND                   | Ground                              |
| 14      | 14         | DL2p                  | Receiver non-inverted data output   |
| 15      | 15         | DL2n                  | Receiver inverted data output       |
| 16      | 16         | GND                   | Ground                              |
| 17      | 17         | DL0p                  | Receiver non-inverted data output   |
| 18      | 18         | DL0n                  | Receiver inverted data output       |
| 19      | 19         | GND                   | Ground                              |
| 20      | 20         | GND                   | Ground                              |
| 21      | 21         | DL1n                  | Receiver inverted data output       |
| 22      | 22         | DL1p                  | Receiver non-inverted data output   |
| 23      | 23         | GND                   | Ground                              |
| 24      | 24         | DL3n                  | Receiver inverted data output       |
| 25      | 25         | DL3p                  | Receiver non-inverted data output   |
| 26      | 26         | GND                   | Ground                              |
| 32      | 32         | GND                   | Ground                              |
| 33      | 33         | SL2p                  | Transmitter non-inverted data input |
| 34      | 34         | SL2n                  | Transmitter inverted data input     |
| 35      | 35         | GND                   | Ground                              |
| 36      | 36         | SL0p                  | Transmitter non-inverted data input |
| 37      | 37         | SL0n                  | Transmitter inverted data input     |
| 38      | 38         | GND                   | Ground                              |
| —       | 39         | GND                   | Ground                              |
| —       | 40         | SL5n                  | Transmitter inverted data input     |
| —       | 41         | SL5p                  | Transmitter non-inverted data input |

**Table 162C–4—MDI connector contact mapping for QSFP112 and QSFP-DD800 (*continued*)**

| QSFP112 | QSFP-DD800 | Connector signal name | Description                         |
|---------|------------|-----------------------|-------------------------------------|
| —       | 42         | GND                   | Ground                              |
| —       | 43         | SL7n                  | Transmitter inverted data input     |
| —       | 44         | SL7p                  | Transmitter non-inverted data input |
| —       | 45         | GND                   | Ground                              |
| —       | 51         | GND                   | Ground                              |
| —       | 52         | DL6p                  | Receiver non-inverted data output   |
| —       | 53         | DL6n                  | Receiver inverted data output       |
| —       | 54         | GND                   | Ground                              |
| —       | 55         | DL4p                  | Receiver non-inverted data output   |
| —       | 56         | DL4n                  | Receiver inverted data output       |
| —       | 58         | GND                   | Ground                              |
| —       | 59         | DL5n                  | Receiver inverted data output       |
| —       | 60         | DL5p                  | Receiver non-inverted data output   |
| —       | 61         | GND                   | Ground                              |
| —       | 62         | DL7n                  | Receiver inverted data output       |
| —       | 63         | DL7p                  | Receiver non-inverted data output   |
| —       | 64         | GND                   | Ground                              |
| —       | 70         | GND                   | Ground                              |
| —       | 71         | SL6p                  | Transmitter non-inverted data input |
| —       | 72         | SL6n                  | Transmitter inverted data input     |
| —       | 73         | GND                   | Ground                              |
| —       | 74         | SL4p                  | Transmitter non-inverted data input |
| —       | 75         | SL4n                  | Transmitter inverted data input     |

**Table 162C–5—MDI connector contact mapping for OSFP**

| OSFP | Connector<br>signal name | Description                         |
|------|--------------------------|-------------------------------------|
| 1    | GND                      | Ground                              |
| 2    | SL1p                     | Transmitter non-inverted data input |
| 3    | SL1n                     | Transmitter inverted data input     |
| 4    | GND                      | Ground                              |
| 5    | SL3p                     | Transmitter non-inverted data input |
| 6    | SL3n                     | Transmitter inverted data input     |
| 7    | GND                      | Ground                              |
| 8    | SL5p                     | Transmitter non-inverted data input |
| 9    | SL5n                     | Transmitter inverted data input     |
| 10   | GND                      | Ground                              |
| 11   | SL7p                     | Transmitter non-inverted data input |
| 12   | SL7n                     | Transmitter inverted data input     |
| 13   | GND                      | Ground                              |
| 18   | GND                      | Ground                              |
| 19   | DL6n                     | Receiver inverted data output       |
| 20   | DL6p                     | Receiver non-inverted data output   |
| 21   | GND                      | Ground                              |
| 22   | DL4n                     | Receiver inverted data output       |
| 23   | DL4p                     | Receiver non-inverted data output   |
| 24   | GND                      | Ground                              |
| 25   | DL2n                     | Receiver inverted data output       |
| 26   | DL2p                     | Receiver non-inverted data output   |
| 27   | GND                      | Ground                              |
| 28   | DL0n                     | Receiver inverted data output       |
| 29   | DL0p                     | Receiver non-inverted data output   |
| 30   | GND                      | Ground                              |
| 31   | GND                      | Ground                              |
| 32   | DL1p                     | Receiver non-inverted data output   |
| 33   | DL1n                     | Receiver inverted data output       |
| 34   | GND                      | Ground                              |
| 35   | DL3p                     | Receiver non-inverted data output   |
| 36   | DL3n                     | Receiver inverted data output       |

**Table 162C–5—MDI connector contact mapping for OSFP (*continued*)**

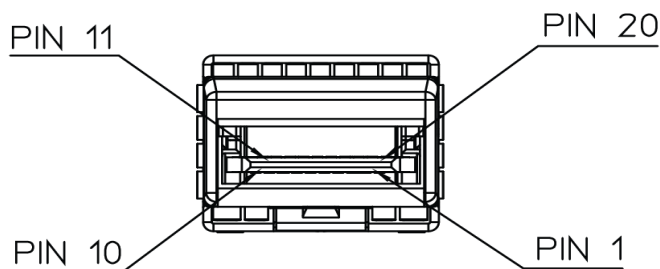
| OSFP | Connector<br>signal name | Description                         |
|------|--------------------------|-------------------------------------|
| 37   | GND                      | Ground                              |
| 38   | DL5p                     | Receiver non-inverted data output   |
| 39   | DL5n                     | Receiver inverted data output       |
| 40   | GND                      | Ground                              |
| 41   | DL7p                     | Receiver non-inverted data output   |
| 42   | DL7n                     | Receiver inverted data output       |
| 43   | GND                      | Ground                              |
| 48   | GND                      | Ground                              |
| 49   | SL6n                     | Transmitter inverted data input     |
| 50   | SL6p                     | Transmitter non-inverted data input |
| 51   | GND                      | Ground                              |
| 52   | SL4n                     | Transmitter inverted data input     |
| 53   | SL4p                     | Transmitter non-inverted data input |
| 54   | GND                      | Ground                              |
| 55   | SL2n                     | Transmitter inverted data input     |
| 56   | SL2p                     | Transmitter non-inverted data input |
| 57   | GND                      | Ground                              |
| 58   | SL0n                     | Transmitter inverted data input     |
| 59   | SL0p                     | Transmitter non-inverted data input |
| 60   | GND                      | Ground                              |

## 162C.2 MDI connector types

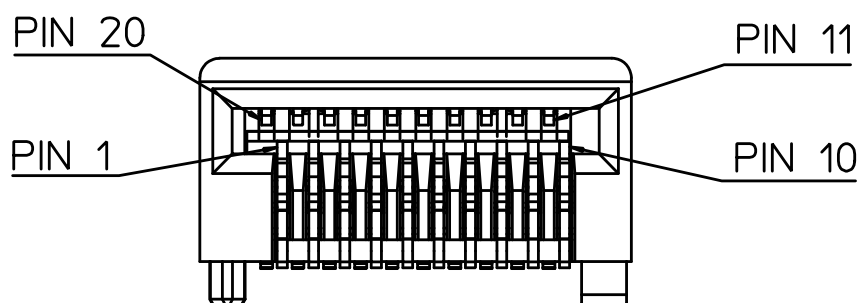
### 162C.2.1 SFP112

For 100GBASE-CR1, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of the SFP-DD MSA. SFP112 supports one lane.

The SFP112 cable assembly connector is the plug illustrated in Figure 162C–1. The SFP112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C–2. The connectors have data signal and signal ground contact assignments as specified in Table 162C–3.



**Figure 162C-1—SFP112 cable assembly plug**

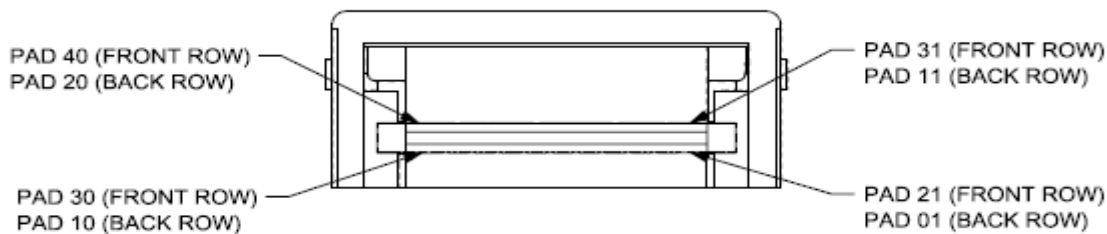


**Figure 162C-2—SFP112 PMD receptacle**

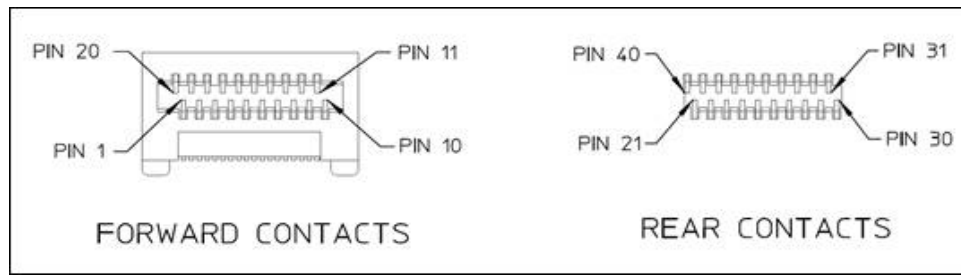
### 162C.2.2 SFP-DD112

For 100GBASE-CR1 or 200GBASE-CR2 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the SFP-DD MSA. SFP-DD112 supports up to two lanes.

The SFP-DD112 cable assembly connector is the plug illustrated in Figure 162C-3. The SFP-DD112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-4. The connectors have data signal and signal ground contact assignments as specified in Table 162C-3.



**Figure 162C-3—SFP-DD112 cable assembly plug**

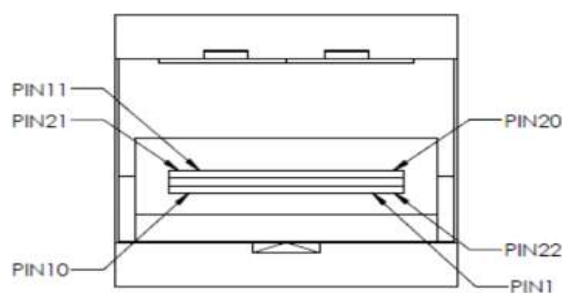


**Figure 162C-4—SFP-DD112 PMD receptacle**

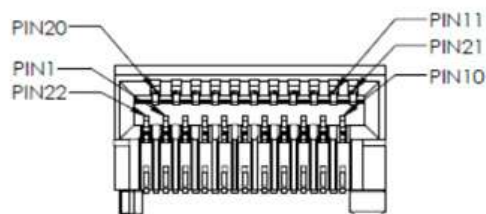
### 162C.2.3 DSFP

For 100GBASE-CR1 or 200GBASE-CR2 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the DSFP MSA. DSFP supports up to two lanes.

The DSFP cable assembly connector is the plug illustrated in Figure 162C-5. The DSFP PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-6. The connectors have data signal and signal ground contact assignments as specified in Table 162C-3.



**Figure 162C-5—DSFP cable assembly plug**



**Figure 162C-6—DSFP PMD receptacle**

## 162C.2.4 QSFP112

For 100GBASE-CR1, 200GBASE-CR2 or 400GBASE-CR4 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of the QSFP-DD MSA. QSFP112 supports up to four lanes.

The QSFP112 cable assembly connector is the plug illustrated in Figure 162C-7. The QSFP112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-8. The connectors have data signal and signal ground contact assignments as specified in Table 162C-4.

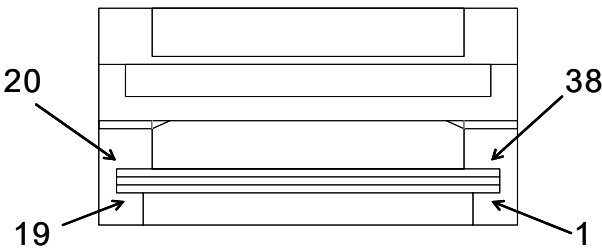


Figure 162C-7—QSFP112 cable assembly plug

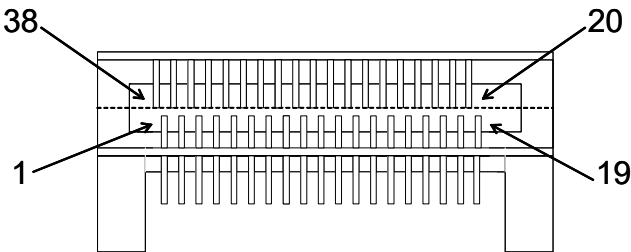


Figure 162C-8—QSFP112 PMD receptacle

162C.2.5 QSFP-DD800

For 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined as defined in the QSFP-DD800 MSA. QSFP-DD800 supports up to eight lanes.

The QSFP-DD800 cable assembly connector is the plug illustrated in Figure 162C-9. The QSFP-DD PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-10. The connectors have data signal and signal ground contact assignments as specified in Table 162C-4.

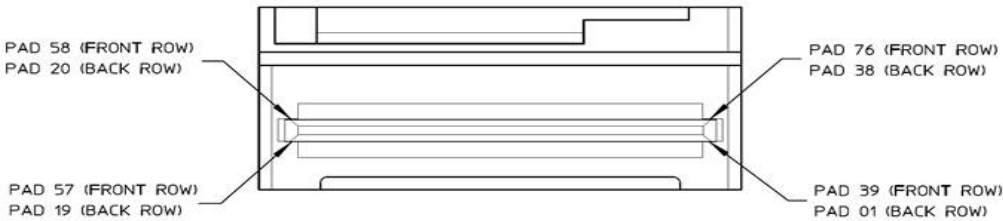


Figure 162C-9—QSFP-DD800 cable assembly plug

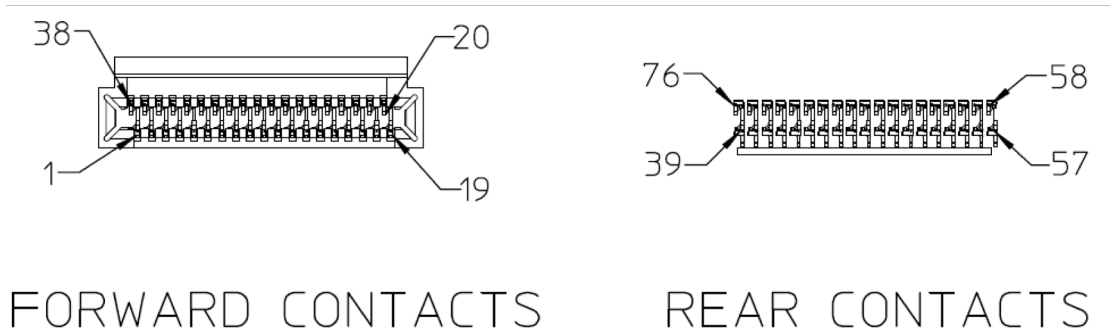


Figure 162C-10—QSFP-DD800 PMD receptacle

### 162C.2.6 OSFP

For 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the OSFP MSA. OSFP supports up to eight lanes.

The OSFP cable assembly connector is the plug illustrated in Figure 162C-11. The OSFP PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-12. The connectors have data signal and signal ground contact assignments as specified in Table 162C-5.

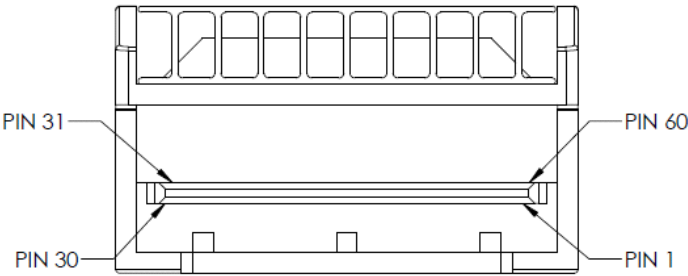
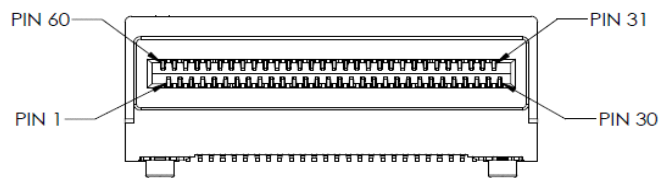


Figure 162C-11—OSFP cable assembly plug



**Figure 162C–12—OSFP PMD receptacle**

## 162C.3 Protocol implementation conformance statement (PICS) proforma for Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4<sup>27</sup>

### 162C.3.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

### 162C.3.2 Identification

#### 162C.3.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                    |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                  |  |
| Implementation Name(s) and Version(s) <sup>1, 3</sup>                                                                                                                                                                                                                                    |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                      |  |
| NOTE 1—Required for all implementations.<br>NOTE 2—May be completed as appropriate in meeting the requirements for the identification.<br>NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model). |  |

#### 162C.3.2.2 Protocol summary

|                                                                                                                                                                                                                               |                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                                                                           | IEEE Std 802.3ck-2022, Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                                                                               |                                                                                          |
| Have any Exception items been required? No <input type="checkbox"/> Yes <input type="checkbox"/><br>(See <a href="#">Clause 21</a> ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.) |                                                                                          |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>27</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

### 162C.3.3 Major capabilities/options

| Item  | Feature                  | Subclause | Value/Comment                                      | Status | Support           |
|-------|--------------------------|-----------|----------------------------------------------------|--------|-------------------|
| *MDI1 | SFP112 MDI connector     | 162C.2.1  | Device uses SFP112 receptacle as MDI connector     | O      | Yes [ ]<br>No [ ] |
| *MDI2 | SFP-DD112 MDI connector  | 162C.2.2  | Device uses SFP-DD112 receptacle as MDI connector  | O      | Yes [ ]<br>No [ ] |
| *MDI3 | DSFP MDI connector       | 162C.2.3  | Device uses DSFP receptacle as MDI connector       | O      | Yes [ ]<br>No [ ] |
| *MDI4 | QSFP112 MDI connector    | 162C.2.4  | Device uses QSFP112 receptacle as MDI connector    | O      | Yes [ ]<br>No [ ] |
| *MDI5 | QSFP-DD800 MDI connector | 162C.2.5  | Device uses QSFP-DD800 receptacle as MDI connector | O      | Yes [ ]<br>No [ ] |
| *MDI6 | OSFP MDI connector       | 162C.2.6  | Device uses OSFP receptacle as MDI connector       | O      | Yes [ ]<br>No [ ] |

### 162C.3.4 PICS proforma tables for MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

#### 162C.3.4.1 Contact Mapping

| Item | Feature                                                            | Subclause | Value/Comment | Status | Support            |
|------|--------------------------------------------------------------------|-----------|---------------|--------|--------------------|
| CM1  | SFP112 MDI connector contact mapping according to Table 162C–3     | 162C.2.1  |               | MDI1:M | Yes [ ]<br>N/A [ ] |
| CM2  | SFP-DD112 MDI connector contact mapping according to Table 162C–3  | 162C.2.2  |               | MDI2:M | Yes [ ]<br>N/A [ ] |
| CM3  | DSFP MDI connector contact mapping according to Table 162C–3       | 162C.2.3  |               | MDI3:M | Yes [ ]<br>N/A [ ] |
| CM4  | QSFP112 MDI connector contact mapping according to Table 162C–4    | 162C.2.4  |               | MDI4:M | Yes [ ]<br>N/A [ ] |
| CM5  | QSFP-DD800 MDI connector contact mapping according to Table 162C–4 | 162C.2.5  |               | MDI5:M | Yes [ ]<br>N/A [ ] |
| CM6  | OSFP MDI connector contact mapping according to Table 162C–5       | 162C.2.6  |               | MDI6:M | Yes [ ]<br>N/A [ ] |